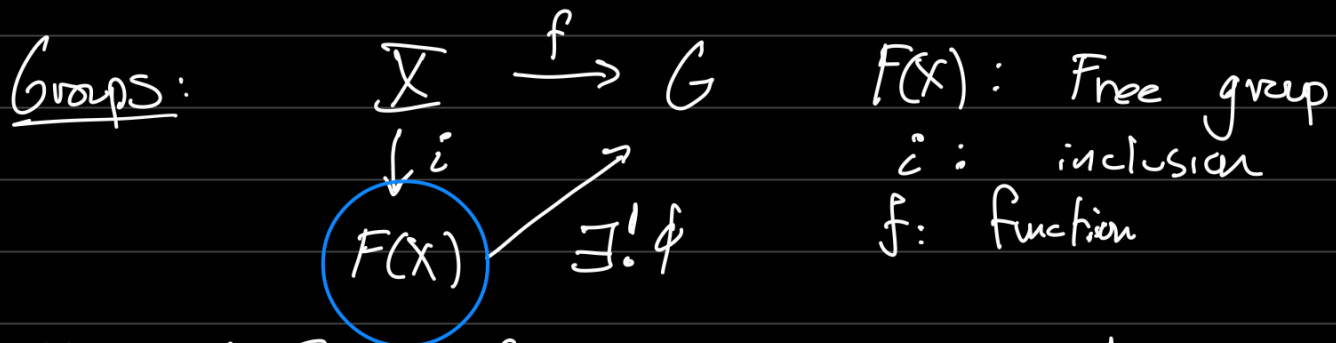


* Universal property: characterizes an object by how it relates to all other objects through morphisms

Category theory:

Concept	Description	Ex
Object :	Structure	Set, group, Vec. space
Morphism :	Structure preserving map	f^n , $\hat{\text{hom}}$, linear map
Category :	collection of obj + morph.	<u>Set</u> , <u>Grp</u> , <u>Vect</u>
Functor :	Map between Categories	Forgetful, free, dual
Natural transformation:	Map between functors	Determinant

Free group: $F(X)$, $x_i \in X$
 $\varepsilon_i \in \{+1, -1\}$
 $F(X) \equiv \{ x_{i_1}^{\varepsilon_1} \dots x_{i_n}^{\varepsilon_n} \mid \}$



Claim: $\phi: F(X) \rightarrow G$ is unique group homo.

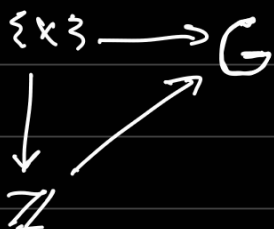
or

$$\begin{array}{l}
 \phi \circ i = f \\
 \phi|_X = f
 \end{array}$$

$F(X)$: Most general group containing X

ex: $F(\{x\}) = (\mathbb{Z}, +)$

$$X = \{x\}, \quad F(\{x\}) = \mathbb{Z}$$



Quotient groups: $N \trianglelefteq G$: N is a normal subgroup of G

H is a subgroup of G $f: G \rightarrow H \leq \text{Per}(M)$

$$\Rightarrow \exists! \bar{f} : G/N \rightarrow H$$

$$G \xrightarrow{f} H \quad \bar{f} \circ \pi = f$$

* G/N : Most general group "without N " $\nwarrow$ Binary op.

Monoids: a set with an associative operation + identity

* Free Monoids: is a monoid

$$X \xrightarrow{\quad} N$$